

CRIMECAST

How the Project Works — Simple English

What it does · Modules · Algorithms · Data · Tools

College prototype · Not police system · Not official SCRB forecast

30-second summary: CRIMECAST cleans multi-year TN crime CSVs, trains classical ML (Ridge, Random Forest, Gradient Boosting), predicts with history blend, uses news for live context (not FIRs), and shows 2026 as a **scenario** with maps, compare, and simple explanations.

1. What is this project?

CRIMECAST is a student dashboard that helps people look at district crime numbers, make ML predictions, sketch a 2026 scenario, and see news sentiment for Tamil Nadu.

- Look at crime-related numbers by district
- Predict rates/counts with machine learning
- Sketch 2026 scenarios (not official forecasts)
- See news headlines and sentiment
- Use one Streamlit website for maps and compare

Not a live police system. **Not** an official government forecast.

2. What does it do?

Feature	In simple words
Load crime tables	Reads CSVs (complaints, murder, crimes against women)
Clean data	Fixes names, removes TOTAL rows, standardises districts
Train models	Learns patterns from past official numbers
Predict	Estimates a district value for a chosen year
2026 scenarios	Simple trends + uncertainty range
News + sentiment	Headlines more positive or negative
Dashboard	Maps, lists, compare, explain in browser
Tests	Automated checks via run_tests.py

3. How it works (pipeline)

Data → **Clean** → **Train** → **Predict** → **News/Sentiment** → **Dashboard** (+ 2026 trend branch)

- DATA — Official CSVs + news headlines
- CLEAN — One ML-ready table; TN38 district names
- TRAIN — Ridge / RF / GB; pick best per target
- PREDICT — Model + district history blend
- FORECAST — Linear / last-year / blend for 2026
- NEWS — Headlines → sentiment / map heat
- DASHBOARD — Streamlit UI for the user

Key idea: Official tables train models. News is for live awareness only.

4. Modules (main files)

File	What it does
clean_data.py	Clean CSVs → ML-ready table
train_model.py	Train models; save .joblib
predict.py	Predict + history blend
forecast_engine.py	2026 multi-target scenarios
sentiment_analysis.py	Text polarity (DistilBERT or lexicon)
acquire_news_signals.py	Refresh media harvest
district_entities.py	City → TN38 parent district
tn_map.py	TN choropleth maps
db.py	SQLite headlines / alerts
dashboard.py	Streamlit website
risk_explain.py	Why high risk (simple + LIME-style)
run_tests.py	Unit / integration / data tests

5. Algorithms (simple English)

5.1 Main ML for crime numbers

Algorithm	Meaning
Dummy (median)	Baseline: always guess the middle value
Ridge Regression	Stable linear model; good for rates
Random Forest	Many decision trees vote
Gradient Boosting	Trees fix errors one after another

Best model chosen using MAE / RMSE / R² and temporal holdout when possible.

5.2 History blend

Final ≈ part model + part past official district value. Rates trust history more; counts trust model more.

5.3 2026 scenario

Method	Meaning
Linear trend	Straight line through past years → 2026
Last year carry	Next looks like last known year
Blend	Average of linear + last year
Uncertainty band	Low / mid / high range

LSTM/Prophet not used for main demo — yearly series are too short.

5.4 Sentiment & explain

DistilBERT (optional) or lexicon keywords; word clouds; composite risk + LIME-style explanations.

6. Data used

Data	Used for
Complaints / murder / women-crime CSVs	Train & predict
Media harvest headlines	Live feed, heat, sentiment
TN38 district list	Maps and fair rollups
Population estimates	Per-lakh ranking
SQLite (optional)	Headlines & alerts store

7. Dashboard screens

Screen	Purpose
Live Feed	Latest news-style items + heat
District Map	TN map + scoreboard
Accuracy	Honest model metrics
Predict	District / target / year → number
Sentiment	Mood map + word cloud
2026 Forecasts	Scenario map + bands
District Compare	2–4 districts side by side
Risk Explain	Why it looks high
Health	Files and models present?

8. Technologies

Python · pandas · numpy · scikit-learn · joblib · Plotly · Streamlit · SQLite · optional transformers/torch

9. What it does not claim

- Not real-time police dispatch
- Not a guaranteed 2026 fact
- News volume is not registered crime
- Short history → scenarios for discussion only

Viva one-liner: We clean multi-table TN crime data, train classical ML models, blend predictions with district history, use news only as support (not FIRs), and show 2026 as a scenario with maps and simple explanations.